

// FEATURE-BY-FEATURE

Capability Teardown

An interactive vector editor, mapped feature by feature against a declarative document format. 51 capabilities, one verdict each — a feature-surface comparison, not a maturity benchmark.

// VERDICT KEY

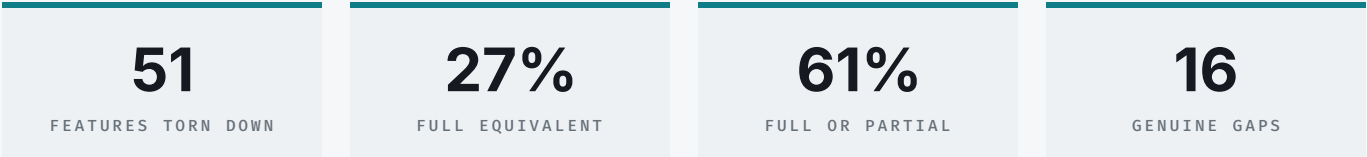
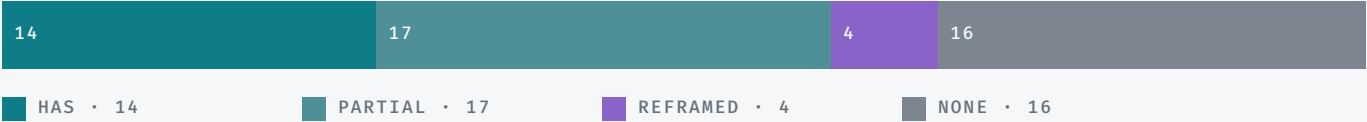
-  HAS a direct equivalent
-  PARTIAL a narrower / different form
-  NONE no equivalent (often a non-goal)
-  REFRAMED exists in both — but reached by code, not cursor

// HOW THIS WAS MEASURED

Method & scorecard

Illustrator's surface was enumerated from three manuals (2024 User's Guide, 2025 book, BMG 106) and reduced to 51 discrete capabilities, each checked against FrameGraph's capability manifest. Evidence cites SRC-ORDINAL. This is a feature-surface comparison, not a maturity benchmark: FrameGraph is a PROPOSED format with no conformant renderer.

COVERAGE ACROSS 51 FEATURES



27% = 14/51 full • 61% = (14+17)/51 full-or-partial (REFRAMED & NONE excluded) • 16 NONE = 10 architectural + 6 declared non-goals •

READING A ROW

AI-nn	monospace feature id (chapter-ordered)
Feature	the Illustrator capability, named
[26•405]	source book (24 / 25 / 26) • sentence ordinal — round-trips
• HAS	FrameGraph verdict glyph + label
evidence	the FrameGraph capability that backs the verdict



WHAT “REFRAMED” MEANS

The capability exists in BOTH tools — but FrameGraph reaches it by naming or a tool call, not a cursor gesture. It is neither a gap nor a look-alike; it is the same end down the opposite (declarative) road.

- AI-01 Illustrator SELECTS an object with the pointer → FrameGraph NAMES it by id.
- AI-40 Illustrator TRACES a raster from a menu → FrameGraph calls the vectorize_image tool.

// HOW A VERDICT IS DECIDED

Rubric, sources & audit identity

VERDICT RUBRIC — THE THRESHOLD FOR EACH LABEL

 HAS	A direct FUNCTIONAL equivalent: the same user outcome is expressible, even if authored declaratively.
 PARTIAL	Narrower or different: missing interactivity, missing output fidelity, or a workflow-only (not functional) match.
 REFRAMED	Exists in both, but reached by NAMING / a tool call / an author→render loop — not a cursor. Not a gap; a different road.
 NONE	No equivalent. Tagged by gap_type: non_goal (scope choice) · arch (declarative model precludes it) · maturity (plausible, unbuilt).

CONFIDENCE (SHOWN ON M / L ROWS)

H	high — direct, well-evidenced
M	medium — interpretive mapping
L	low — loose equivalence, flagged for review

EQUIVALENCE DIMENSION

F	functional — same user outcome
W	workflow — same task, different steps
D	data-model — structural analogue only

SCORE MATH (NO WEIGHTING)

Full equivalent	has/total = 14/51
Full or partial	(has+part)/total = 31/51 (REFRAMED & NONE excluded)
Reachable by any route	(has+part+reframed)/total = 35/51 (REFRAMED counted as reachable)

SOURCES (DOC-RAY CORPUS · GRAPHQL)

[24] Adobe Illustrator 2024 User's Guide — 231 pp, 2,283 sentences.
[25] Master Adobe Illustrator 2025 — Dana J. Bailey, 159 pp.
[26] BMG 106: Computer Graphics II — Adobe Illustrator,
Each [SRC·ORDINAL] round-trips via searchSentences().

AUDIT IDENTITY

FrameGraph	v2.3.0
Manifest	docs/capability-manifest.json
Capabilities	278
Manifest sha256	51b966f9c609ff1e913885ff...
Full audit	illustrator_vs_framegraph.audit.json
Gate	tests/test_capability_manifest.py

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// THE TEARDOWN

Feature-by-feature

A • SELECT & EDIT PATHS

ID	ILLUSTRATOR FEATURE • EVIDENCE [SRC•ORDINAL]	FG	FRAMEGRAPH COVERAGE
AI-01	Object selection Selection Tool picks a whole object to move or transform. [25•64]	 REFRAMED • H	REFRAMED: you NAME an object by id — declaration replaces cursor selection.
AI-02	Anchor-point editing Direct Selection edits specific anchor points & paths. [25•198]	 PARTIAL • M	path / bezier / curve points authored as coordinates.
AI-03	Isolation mode Double-click into a group to edit its contents in place. [25•281]	 PARTIAL • M	structural nesting (group + layers) — no in-place isolated edit.
AI-04	Compound paths / Pathfinder Combine paths (Pathfinder) so one cuts a hole in another. [24•1488]	 PARTIAL • M	path fill_rule even-odd + holes; no live Pathfinder ops.
AI-05	Shape Builder Drag across shapes to merge / subtract them interactively. [24•523]	 NONE • H architectural	no boolean / interactive shape merge.
AI-06	Scissors & Knife Cut a path at a point, or slice across objects. [25•365]	 NONE • H architectural	no interactive path surgery.
AI-47	Offset path Object > Path > Offset expands a path outward / inward. [26•3298]	 NONE • H architectural	no offset-path operation.
AI-48	Outline stroke Object > Path > Outline Stroke converts a stroke to a filled path. [26•771]	 NONE • H architectural	no stroke→outline conversion.

// THE TEARDOWN, CONT.

Feature-by-feature

B • DRAW & PRIMITIVES

ID	ILLUSTRATOR FEATURE • EVIDENCE [SRC•ORDINAL]	FG	FRAMEGRAPH COVERAGE
AI-07	Shape primitives Rectangle, Ellipse, Polygon, Star, Line, Arc tools. [25•47]	 HAS	rect, ellipse, circle, polygon, line, polyline (17 object_types).
AI-08	Pen tool (Bézier) Set Bézier anchor points and handles by hand. [24•612]	 PARTIAL • M	bezier / Path / CubicBezier objects, authored by coordinate.
AI-09	Curvature tool Rubber-band free-form curves as you click & release. [24•1964]	 PARTIAL • M	curve object + parametric_curve; no live rubber-band.
AI-10	Pencil / freehand Sketch a path freehand; Illustrator fits the curve. [24•747]	 NONE • H non-goal	no freehand pointer input (declarative only).
AI-11	Stroke controls Weight, dashes, caps, joins, arrowheads on the Stroke panel. [25•93]	 HAS	stroke_style: width / dasharray / cap / join + connector markers.
AI-12	Variable-width (Width Tool) Vary stroke width along a path for a calligraphic look. [24•666]	 NONE • H architectural	stroke width is uniform — no width profile.
AI-49	Brushes Calligraphic, Scatter, Art, Pattern, Blob brushes paint along a path. [26•937]	 NONE • H architectural	no brush engine; nearest is hatch_fill / pattern.

// THE TEARDOWN, CONT.

Feature-by-feature

C • COLOUR SYSTEM

ID	ILLUSTRATOR FEATURE • EVIDENCE [SRC•ORDINAL]	FG	FRAMEGRAPH COVERAGE
AI-13	Colour picker & swatches Pick colour; save reusable named swatches. [25•85]	 HAS	hex / rgba values + defs.tokens.colors named palette.
AI-14	Global colours Edit a global swatch once; every use updates. [25•958]	 HAS	named colour tokens are global by construction.
AI-15	CMYK / RGB modes Document colour mode selectable CMYK or RGB. [25•326]	 PARTIAL • H	RGB / hex only — no CMYK, no separations.
AI-16	Recolor Artwork Remap a whole palette via the Recolor command / Color Guide. [24•684]	 PARTIAL • M	gradient_map + closed_palette + token swap (declarative recolor).
AI-17	Live Paint Fill regions bounded by overlapping paths interactively. [24•319]	 PARTIAL • M	region toolkit: select_in / place_region / region_grade.
AI-18	Colour science Colour groups & harmony via the Color Guide panel. [24•684]	 PARTIAL • M	Chevreur harmony + WCAG contrast tools — not a Color-Guide workflow.
AI-19	Patterns Tile a swatch as a repeating pattern fill. [24•695]	 HAS	pattern, grid_pattern, dots, hatch_fill fills.

// THE TEARDOWN, CONT.

Feature-by-feature

D • TYPE & TEXT

ID	ILLUSTRATOR FEATURE • EVIDENCE [SRC•ORDINAL]	FG	FRAMEGRAPH COVERAGE
AI-20	Point & area type Type tool sets point text or text inside a shape. [25•1015]	 HAS	text objects + flow paragraph / heading / list.
AI-21	Character & paragraph Font, size, weight, alignment, spacing, indents. [25•1029]	 HAS	font_family / size / weight + paragraph flow (106 style props).
AI-22	Threaded text Flow overset text between linked frames. [25•1029]	 PARTIAL • M	flow auto-pagination continues overset; no interactive frame linking.
AI-23	Type on a path Set text running along a curved path. [25•1099]	 NONE • H non-goal	no text-on-path (explicit scope limit).
AI-24	Kerning & tracking Fine letter / line spacing on the Character panel. [25•1054]	 PARTIAL • M	letter_spacing / line_height; no pair-kerning table.
AI-25	Envelope distort Warp text into an arbitrary envelope shape. [25•715]	 NONE • H non-goal	non-goal — text stays on its baseline grid.

// THE TEARDOWN, CONT.

Feature-by-feature

E • GRADIENT • EFFECT • STYLE

ID	ILLUSTRATOR FEATURE • EVIDENCE [SRC•ORDINAL]	FG	FRAMEGRAPH COVERAGE
AI-26	Linear / radial gradient Colour stops along a linear or radial axis. [24•844]	 HAS	linear_gradient / radial_gradient paint.
AI-27	Freeform gradient Colour stops placed freely across the object. [24•857]	 NONE • H architectural	no freeform gradient.
AI-28	Gradient mesh A mesh of colour points for painterly blends. [24•1079]	 NONE • H architectural	no mesh — the single biggest gap.
AI-29	Blend tool Interpolate shape & colour between two objects. [24•1160]	 NONE • H architectural	no shape-to-shape blend / interpolation.
AI-30	Live effects Drop shadow, glow, blur — non-destructive effects. [25•1496]	 PARTIAL • M	effects set: glow / neon / shadow / soft_shadow / hatch.
AI-31	Graphic styles Save an appearance and reapply it by name. [24•1823]	 HAS	named styles / text_styles / stroke_styles tokens.
AI-32	Appearance stack Stack multiple fills / strokes / effects per object. [24•1092]	 PARTIAL • M	one style per object; no multi-fill appearance stack.

// THE TEARDOWN, CONT.

Feature-by-feature

F • LAYER • TRANSFORM • PAGE

ID	ILLUSTRATOR FEATURE • EVIDENCE [SRC·ORDINAL]	FG	FRAMEGRAPH COVERAGE
AI-33	Layers Organise artwork in a Layers-panel hierarchy. [24·547]	 HAS	ordered layers per page, z-index.
AI-34	Transform tools Rotate, Scale, Reflect, Shear an object. [24·700]	 HAS	transform: Mat3 rotate / scale / translate / shear.
AI-35	Align & distribute Align / distribute objects on the Align panel. [24·542]	 HAS	layout groups: row / column / grid, align.
AI-36	Artboards Multiple artboards — “similar to pages in InDesign”. [25·423]	 PARTIAL • M	multi-page flow doc (TOC, bibliography); no free spatial canvas.
AI-37	Perspective grid Draw & place objects on a simulated perspective grid. [24·762]	 NONE • H non-goal	no perspective grid (flat page space).
AI-50	Guides, rulers & snap Rulers, guides and Snap-to-Grid align artwork by eye. [26·572]	 PARTIAL • M	grid_pattern + exact coordinates; no interactive guides / snap.

G • IMAGE • 3D • OUTPUT

ID	ILLUSTRATOR FEATURE • EVIDENCE [SRC·ORDINAL]	FG	FRAMEGRAPH COVERAGE
AI-38	Embed / link raster Embed a raster image, or link it externally. [25·1727]	 HAS	image object: embedded or src-referenced.
AI-39	Image trace Object > Image Trace: raster into editable vectors. [24·1394]	 REFRAMED • H	REFRAMED: same raster→vector via the vectorize_image tool call.
AI-40	3D & Materials Effect > 3D: extrude, bevel, revolve with materials. [24·767]	 PARTIAL • M	Scene3D + Material + Camera scene — not vector extrusion.
AI-41	Export formats EPS, PSD, TIFF, GIF, JPEG, SWF, SVG, DWG/DXF, PDF. [25·20]	 PARTIAL • M	SVG / PNG / PDF / LaTeX (6 renderers); no PSD / EPS / DWG.
AI-42	Package Collect linked assets & fonts into one folder. [24·2204]	 NONE • H non-goal	no asset-collection packaging step.
AI-51	Graph tool Graph tools plot data as bar / pie / line charts. [26·405]	 HAS	Chart / sparkline / function_plot / polar_plot / kpi (data-bound).

// THE TEARDOWN, CONT.

Feature-by-feature

H • GENERATIVE / AI (ILLUSTRATOR 2024)

ID	ILLUSTRATOR FEATURE • EVIDENCE [SRC•ORDINAL]	FG	FRAMEGRAPH COVERAGE
AI-43	Text to Vector Graphic Describe art in words; generative AI makes the vectors. [24•105]	 REFRAMED • L	REFRAMED: propose_from_image / run_sdk_code author→render loop.
AI-44	Generative Recolor Recolour artwork with a text prompt via generative AI. [24•678]	 REFRAMED • L	REFRAMED: swap a palette token / gradient_map, re-render.
AI-45	Retype Identify & match fonts found in existing artwork. [24•73]	 NONE • H architectural	list_fonts resolves families, but no visual font ID.
AI-46	Mockup Wrap flat vector art onto a 3D object mockup. [24•41]	 NONE • H non-goal	no mockup / surface-wrap of art onto an object.

// SHOW, DON'T ASSERT

Every verdict, in source

So the verdicts are not just labels: here is the same intent, as an Illustrator gesture on the left and as the FrameGraph source the SDK emits on the right.

AI-07 · HAS

Draw a rectangle

ILLUSTRATOR

Rectangle Tool → drag on the artboard.

FRAMEGRAPH SDK

```
page.rect([40, 40, 220, 120],
fill="#1f9ec4",
**stroke(2, color="ink"))
```

AI-01 · REFRAMED

Act on one object

ILLUSTRATOR

Selection Tool → click the object → move.

FRAMEGRAPH SDK

```
# no cursor — name it, then act
card = page.rect([...], id="hero")
page.move("hero", dx=12, dy=0)
```

AI-26 · HAS

Linear gradient fill

ILLUSTRATOR

Gradient panel → set stops → drag.

FRAMEGRAPH SDK

```
page.rect([40, 40, 300, 80], fill=linear_gradient(
stops=[(0, "#0f7d88"), (1, "#f5f7f9")], angle=0))
```

AI-02 · PARTIAL · M

Edit an anchor point

ILLUSTRATOR

Direct Selection → drag a point by hand.

FRAMEGRAPH SDK

```
# authored by coordinate, not dragged
page.path([[40,40],[120,20],[200,60]])
# to 'edit' AI-02: change the numbers
```

AI-28 · NONE (arch)

Gradient mesh

ILLUSTRATOR

Mesh Tool → add colour points → blend.

FRAMEGRAPH SDK

```
# no mesh primitive in the model —
# nearest: a linear/radial gradient,
# which cannot vary in two axes.
```

// THE COMPARISON RUNS BOTH WAYS

What FrameGraph has that Illustrator lacks

A comparison that only lists an editor's features would be rigged. These are manifest capabilities a vector editor has no NATIVE, declarative answer for — the price Illustrator pays for being a canvas, not a document engine. (Illustrator can still embed such artwork by hand, via scripts or plugins; it just is not native.)

◆ Long-form documents

flow model: TOC, bibliography, footnotes, code & math blocks, keep_together — a book, not an artboard.

◆ Structural validation

a schema + validator + golden-render lock prove the document before a pixel is drawn; Illustrator has no correctness gate.

◆ Reproducible source of truth

plain-text YAML, diffable & version-controlled, with byte-stable renders — not a binary .ai project.

◆ Parametric geometry

parametric_curve, Lattice, manifolds (Klein bottle, Möbius) — equation-driven geometry Illustrator has no tool for. (Charts it does have: AI-51.)

◆ UI component library

navbar, button, card, tabs, badge, avatar, breadcrumb, slider — a whole interface kit with no Illustrator equivalent.

◆ Colour science

Chevreul harmony + WCAG contrast_ratio + tone_scale — measured colour, not just swatches.

◆ Machine authoring

25 MCP tools: an AI author→render→verify loop. The document is built and checked by code, end to end.

// WHAT THE TALLY MEANS

Two machines, one honest score

Across 51 Illustrator capabilities from three manuals, FrameGraph has a full equivalent for 14 and a partial one for 17. The 16 genuine gaps cluster where Illustrator is a painting instrument — gradient mesh, freeform gradient, blends, the Pencil, variable-width strokes, perspective, envelope distort, mockup. The 4 REFRAMED rows are the most telling: selection, image trace, and the new 2024 generative-AI features exist in both — but FrameGraph reaches them by naming, a tool call, and an author→render loop, not a cursor.

Illustrator is a hand.
FrameGraph is a grammar.

The gaps are the price of being declarative and machine-authored;
the overlaps are where a format can meet an editor.

// HONEST SCOPE

FrameGraph v2 is a PROPOSED format: no renderer is conformant, the SVG and matplotlib backends are sanity checks, not fidelity guarantees. Illustrator is a mature product. This is a comparison of feature surface and design intent, not of production maturity.

// PROVENANCE

Illustrator features mined from THREE doc-ray corpus sources over GraphQL: [24] Adobe Illustrator 2024 User's Guide (231 pp), [25] Master Adobe Illustrator 2025 (Dana J. Bailey, 159 pp), and [26] BMG 106: Computer Graphics II (257 pp). Each SRC·ORDINAL round-trips via searchSentences(). FrameGraph coverage quotes docs/capability-manifest.json (278 capabilities, tri-state core/SDK/MCP), gated by tests/test_capability_manifest.py.

Composed by Claude (Fable 5), 2026 · a declarative teardown of a declarative format.